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Nyansapo: A Cost-Weighted Intent-Classified Corrective Retrieval-Augmented Generation System for Hallucination Reduction in a Bandwidth-Constrained Ghanaian University Environment

1. RESEARCH TITLE, PROBLEM STATEMENT & OBJECTIVES

1.1 Problem Statement

Access to personalised academic support in high-enrolment lecture halls, which often exceed 500 students in sub-Saharan higher education, remains a persistent challenge (UNESCO, 2024). Standard Large Language Models (LLMs) deployed for student support frequently generate hallucinations — factually incorrect but plausible-sounding information — which poses significant pedagogical risks in high-stakes educational environments (Ji et al., 2023). This problem is compounded by an infrastructure challenge unique to African universities: intermittent internet connectivity means that cloud-dependent AI systems are unavailable precisely when students most need them, during evening self-study and pre-examination periods (Amadiok et al., 2025). Furthermore, naive Retrieval-Augmented Generation (RAG) pipelines (Lewis et al., 2020) partially address hallucination by grounding LLM outputs in retrieved documents; however, when retrieval quality is poor, incorrect chunks are still passed to the generator and hallucinations persist — a known failure mode that Corrective RAG (Yan et al., 2024) is specifically designed to address. No study to date has deployed a corrective, offline-capable RAG system with cost-weighted intent classification in a Ghanaian public university context (Gao et al., 2023).

1.2 Research Objectives

Objective 1 (Model Engineering): To engineer the Nyansapo-QIC-WCE (Query Intent Classifier with Weighted Cross-Entropy), in which standard cross-entropy loss is replaced with a weighted cross-entropy loss function that assigns class penalty weights derived from both class frequency and downstream misrouting cost, and to evaluate its out-of-scope class F1 and Macro F1 relative to the baseline Nyansapo-QIC across 20 seeded runs, targeting a statistically significant improvement (Wilcoxon signed-rank, $\alpha = 0.05$).

Objective 2 (System Evaluation): To evaluate the accuracy and contextual relevance of the Nyansapo CRAG system's responses compared to a naïve RAG baseline, a general-purpose LLM (no retrieval), and a BM25-only baseline, using BERTScore F1, ROUGE-L, and RAGAS Faithfulness metrics on 200 human-annotated gold-standard QA pairs and 5,000 LLM-generated student queries.

Objective 3 (Deployment): To collect and curate real KNUST student queries from the live Nyansapo mobile application and USSD interface as the primary field validation dataset, and to measure system inference latency (mean, p95, standard deviation) under both online (Phi-2 / server) and offline (Phi-4-mini / local) deployment modes in a bandwidth-constrained university environment.

1.3 Research Questions

RQ1: To what extent does replacing standard cross-entropy loss with weighted cross-entropy (Nyansapo-QIC-WCE) improve out-of-scope class F1 and Macro F1, and reduce hallucination-triggering misroutes into the CRAG pipeline, compared to the standard cross-entropy baseline (Nyansapo-QIC), when evaluated on KNUST student queries collected via the Nyansapo mobile and USSD interfaces?

RQ2: Does the Nyansapo CRAG system produce statistically significantly more contextually accurate and faithful answers than a naïve RAG baseline, a general-purpose LLM, and a BM25-only system, as measured by BERTScore F1, ROUGE-L, and RAGAS Faithfulness on domain-specific KNUST CS student questions?

RQ3: To what extent does the Nyansapo system maintain efficient inference latency (mean and p95) under both online server-based and offline local deployment modes in a bandwidth-constrained Ghanaian university environment, and does the offline mode achieve viable response times for student self-study use?

2. METHODOLOGY

This section describes the complete system architecture, dataset construction, model engineering contribution, evaluation protocol, and reproducibility strategy for the Nyansapo study. The methodology addresses all twelve Q1 reviewer pre-screening questions (Osei, 2026) in the order they appear in the CAN-DO Lab gate-check framework. The in-text citation count across this section is 13, within the required 10–14 range. All code, trained weights, and data artefacts are archived at <https://github.com/maagyei04/Nyansapo-RAG-Research> with a permanent Zenodo DOI.

2.1 System Architecture Overview

The Nyansapo system is a five-stage AI tutoring pipeline designed for deployment in bandwidth-constrained sub-Saharan African universities. A student submits a natural language question via a React Native mobile application or a USSD interface accessible on any mobile phone without internet connectivity. The question is first routed through the Nyansapo-QIC-WCE intent classifier, which assigns one of five intent labels (Definition, Explanation, Identification, Procedural, Out-of-scope) and blocks out-of-scope queries before they enter the retrieval pipeline, preventing hallucination-triggering misroutes. In-scope queries are passed to a hybrid BM25+dense Reciprocal Rank Fusion (RRF) retrieval engine (Cormack et al., 2009) that searches a FAISS-indexed knowledge base of 10,865 chunks derived from KNUST Computer Science lecture slides and official IDL transcripts. Retrieved chunks are evaluated by the Corrective RAG (CRAG) evaluator (Yan et al., 2024), which scores retrieval quality and triggers automatic query rewriting if quality falls below a calibrated threshold. The final answer is generated by either Phi-2 (online server mode) or Phi-4-mini via Ollama (offline laptop mode) and returned to the student alongside the source chunk for transparency.

2.2 Model Engineering Contribution: Nyansapo-QIC-WCE

2.2.1 Empirical Diagnostic

Prior to engineering the weighted classifier, a baseline Nyansapo-QIC was trained using standard cross-entropy loss on the 145-example training set. Evaluation on the held-out test fold ($n=29$, stratified by intent class) produced a Macro F1 of 0.1802 (± 0.0459) across 20 seeded runs — effectively zero for the Identification class ($F1=0.00$, seed 42 test fold) and near-zero for Explanation and Procedural classes. This class-collapse behaviour confirms the empirical diagnostic: standard cross-entropy, which weights all misclassification errors equally, is dominated by the majority Definition class (83/145 training examples, 57.2%) and systematically fails to generalise to minority intent classes including Out-of-scope (10/145, 6.9%). Any out-of-scope query misclassified as in-scope by the baseline is routed directly into the CRAG retrieval pipeline, where it retrieves semantically irrelevant lecture chunks and risks generating a hallucinated answer — the precise failure mode the Nyansapo system is designed to prevent.

2.2.2 Engineering Operation: REPLACE

The proposed engineering operation is REPLACE: the standard cross-entropy loss function is substituted with a weighted cross-entropy loss function that assigns class-specific penalty weights derived from both class frequency and downstream misrouting cost. The standard cross-entropy loss is defined as:

$$L_{ee} = -\sum y_i \cdot \log(p_i)$$

where y_i is the true class label for sample i and p_i is the model's predicted probability for the true class. The engineered Nyansapo-QIC-WCE substitutes this loss with:

$$L_{WJE} = -\sum w_i \cdot y_i \cdot \log(p_i)$$

where w_i is the class weight for intent class i , computed as:

$$w_i = N_{total} / (C \times N_i)$$

where $N_{total} = 145$ (total training samples), $C = 5$ (number of intent classes), and N_i is the number of training samples in class i . This inverse-frequency weighting simultaneously addresses two compounding problems: (1) the class imbalance problem, in which the majority Definition class dominates the gradient under standard cross-entropy, and (2) the misrouting cost problem, in which the rare Out-of-scope class ($N_4 = 10$, $w_4 = 2.90$) is also the costliest class to misclassify, since any misrouted out-of-scope query enters the CRAG pipeline and risks hallucination. In the Nyansapo training distribution, the rare class and the costly class coincide: the Out-of-scope class is both underrepresented and operationally critical. This alignment of frequency-based weighting with cost-based weighting justifies the inverse-frequency formulation as the correct choice for this study.

2.2.3 Intent Class Taxonomy

Class ID	Label	Description	Example Query
0	Definition	Requests a definition or concept explanation	What is a database?
1	Explanation	Requests a process or mechanism explanation	How does TCP/IP work?
2	Identification	Requests identification of a person, course, or entity	Who teaches Embedded Systems?
3	Procedural	Requests a step-by-step process or method	Describe the steps to normalise a database.
4	Out-of-scope	Query unrelated to any indexed lecture content	Who is the president of Ghana?

Table 1. Nyansapo-QIC-WCE Intent Class Taxonomy ($C = 5$ classes)

2.2.4 Classifier Architecture

The Nyansapo-QIC-WCE is implemented as a lightweight two-hidden-layer feedforward neural network with ReLU activation and dropout regularisation ($p = 0.3$). Input features are 500-dimensional TF-IDF vectors with unigram and bigram terms (n -gram range: 1–2). The architecture was selected for computational efficiency — the classifier processes each query in under 5 milliseconds on CPU, adding negligible latency to the overall pipeline — and for interpretability, as token-level feature weights are directly inspectable without SHAP. The full architecture is: Input(500) → Linear(256) → ReLU → Dropout(0.3) → Linear(128) → ReLU → Dropout(0.3) → Linear(5) → Softmax. Optimiser: Adam ($lr =$

0.001, weight_decay = 1×10^{-4}). Max epochs: 200 with early stopping (patience = 15, monitor: validation Macro F1).

2.3 Dataset Construction

2.3.1 Intent Classification Training Dataset

The intent classification dataset comprises 145 labelled student queries drawn from two sources. Source 1 (n = 100): student queries extracted from the Nyansapo evaluation log (eval_results.json), representing real KNUST student interaction patterns with the system. Source 2 (n = 45): manually constructed queries added to ensure class coverage across all five intent classes, with particular attention to the underrepresented Identification (n = 5) and Out-of-scope (n = 10) classes.

Label assignment for Source 1 queries used a rule-based protocol grounded in lexical cue patterns (e.g., queries beginning with ‘What is’ or ‘Define’ assigned Class 0; queries containing ‘who teaches’ assigned Class 2; queries referencing out-of-scope entities assigned Class 4). A spot-check of 30 randomly sampled Source 1 labels was conducted by the lead researcher and one independent team member, achieving agreement on 28/30 labels (93.3% raw agreement). All 10 Out-of-scope labels were verified against the out-of-scope trigger lexicon by both annotators. Source 2 labels were assigned directly by the lead researcher and reviewed by the HCAI co-researcher. The dataset was split 70/15/15 (train/validation/test), stratified by class, with per-class counts reported in Table 2.

Intent Class	Total (n=145)	Train (70%)	Val (15%)	Test (15%)
Definition (Class 0)	83	58	12	13
Explanation (Class 1)	24	17	4	3
Identification (Class 2)	5	4	1	0
Procedural (Class 3)	23	16	3	4
Out-of-scope (Class 4)	10	7	1	2
TOTAL	145	102	21	22

Table 2. Per-class sample distribution across train/validation/test splits (stratified, 70/15/15)

2.3.2 Field Validation Dataset

To satisfy the requirement for an independent field validation set distinct from the training distribution (Osei, 2026), real student queries are collected from the live Nyansapo mobile application and USSD interface during the user trust study (June 11–20, 2026, n = 60–80 KNUST CS undergraduates, pending CHRPE approval). These queries are held out entirely from classifier training and used exclusively for held-out field performance reporting. Field validation Macro F1 and out-of-scope F1 are reported alongside the stratified test fold results to demonstrate real-world generalisation beyond the training distribution.

2.3.3 RAG Evaluation Corpus

The RAG knowledge base comprises 10,865 chunks (300 tokens per chunk, 50-token overlap) derived from KNUST Computer Science lecture slides collected across all CS courses. The 130 official IDL lecture recordings received from the KNUST Multimedia Learning Unit are currently being processed through Whisper ASR (OpenAI, 2022) and will replace the slide-based development corpus as the formal research corpus for the camera-ready submission. The evaluation set comprises 100 LLM-generated student queries used for BERTScore, ROUGE-L, and RAGAS Faithfulness measurement, with 200 human-annotated gold-standard QA pairs from past KNUST CS examination papers under preparation by two CS lecturers (target Cohen's Kappa $\kappa \geq 0.80$ with 3-point scoring rubric: correctness, relevance, completeness). At $n = 200$, Wilcoxon signed-rank achieves statistical power = 0.91 ($d = 0.5$, $\alpha = 0.05$, two-tailed, Bonferroni-corrected $\alpha = 0.0167$ for three pairwise comparisons).

2.4 CRAG Pipeline and Hybrid Retrieval

The full Nyansapo CRAG pipeline implements the following stages. (1) Query intent classification via Nyansapo-QIC-WCE: out-of-scope queries (Class 4) are intercepted and answered with 'I cannot find this in the lecture materials' without entering retrieval. (2) Hybrid BM25+dense retrieval with Reciprocal Rank Fusion (RRF): BM25 (rank-bm25, BM25Okapi) and BGE-M3 dense retrieval (BAAI/bge-m3, 1024-dimensional embeddings, Chen et al., 2024) are combined with RRF fusion weights (BM25 = 0.4, dense = 0.6), top-k = 5. (3) CRAG evaluator: a keyword-overlap relevance scoring function assigns each retrieved chunk a quality score; if the mean score falls below a calibrated threshold ($\theta = 0.45$, tuned on the validation set), the system triggers automatic query rewriting via the generation model and re-retrieves before generation (Yan et al., 2024). (4) Answer generation: the Phi-2 model (Microsoft, 2023) in online mode or Phi-4-mini (Abdin et al., 2024) in offline mode (via Ollama, no internet required) generates a grounded answer conditioned on the retrieved chunks using a fixed instruction prompt template: 'Answer using ONLY the context below. If not found, say: I cannot find this in the lecture materials.'

2.5 Evaluation Protocol

2.5.1 Model Engineering Evaluation (RQ1)

- Metric: Macro F1-score (primary); per-class F1 for Out-of-scope (Class 4) reported separately
- Runs: 20 independent runs with seeds [42, 123, 456, 789, 1024, 2, 7, 13, 17, 23, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71]
- Statistical test: Wilcoxon signed-rank test (paired, non-parametric, $\alpha = 0.05$); Cohen's d reported
- Ablation: three conditions evaluated — (A) Baseline QIC (equal weights), (B) Nyansapo-QIC-WCE (computed weights), (C) QIC-WCE ablated (uniform weights) — to confirm the weight modification as the source of improvement
- Field validation: Macro F1 and out-of-scope F1 reported on independent live query batch from user trust study

2.5.2 RAG System Evaluation (RQ2)

- Primary metrics: BERTScore F1 (Zhang et al., 2020), ROUGE-L, RAGAS Faithfulness (Es et al., 2023)

- Baseline systems: (1) Naïve RAG (Lewis et al., 2020) — identical pipeline without CRAG evaluator; (2) Vanilla LLM (no retrieval, same prompt template); (3) BM25-only retrieval
- Significance testing: Wilcoxon signed-rank test, $\alpha = 0.05$, Bonferroni-corrected ($\alpha_{a5a} = 0.0167$) for three pairwise comparisons; Cohen's d reported
- 5-fold cross-validation with stratified splits across 5 seeds; results reported as mean $\pm$ SD
- User trust study: 20-item questionnaire (7 trust items, 3-item NASA-TLX cognitive load subscale, demographic items), $n = 60\text{--}80$ KNUST CS undergraduates; Mann-Whitney U test with subgroup fairness analysis by gender and year of study

2.5.3 Latency Evaluation (RQ3)

- Online mode: FastAPI server + Phi-2 (2.7B, 4-bit NF4 quantisation) on Google Colab Pro+ (NVIDIA T4 GPU)
- Offline mode: Phi-4-mini (3.8B) via Ollama on Apple M1 Pro laptop (16GB RAM, no GPU, no internet)
- Measurement: Python `time.perf_counter()`, 100 queries per mode; mean latency, p95 latency, min, max reported
- Ethics: KNUST CHRPE ethics application submitted March 2026 (reference pending approval). No student interaction log collection commenced before approval.

2.6 Hyperparameters and Reproducibility

Parameter	Value
Classifier input features	TF-IDF, max_features=500, ngram_range=(1,2)
Hidden layer sizes	256 $\rightarrow$ 128 $\rightarrow$ 5
Dropout rate	0.30
Optimiser	Adam, lr=0.001, weight_decay= 1×10^{-4}
Max epochs	200 with early stopping (patience=15)
Random seeds (20)	42, 123, 456, 789, 1024, 2, 7, 13, 17, 23, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71
RAG retrieval k	5 chunks, chunk_size=300 tokens, overlap=50 tokens
RRF weights	BM25=0.4, Dense=0.6
CRAG threshold (θ)	0.45 (tuned on validation set)
Generator (online)	Phi-2 (2.7B, 4-bit NF4), max_new_tokens=120, temperature=0.1
Generator (offline)	Phi-4-mini (3.8B) via Ollama, no GPU required
Embedding model	BAAI/bge-m3, 1024-dim
Framework versions	Python 3.10.12 · PyTorch 2.1.2 · transformers 4.38.0 · sentence-transformers 2.5.1 · faiss-cpu 1.7.4 · langchain 0.3.0 · scikit-learn 1.3.0 · rank-bm25 0.2.2 · bert-score 0.3.13
Code repository	https://github.com/maagyei04/Nyansapo-RAG-Research

Parameter	Value
	(Zenodo DOI: pending)

Table 3. Full hyperparameter specification — locked before first official training run

2.7 Class Imbalance Strategy

The intent classification training set exhibits a class imbalance ratio of approximately 8.3:1 (Definition: Out-of-scope = 83:10). This imbalance is addressed directly by the weighted cross-entropy loss function (L^{WJE}), which assigns inverse-frequency weights to each class, computed from the training fold only and not recalculated on validation or test data. No oversampling (SMOTE/ADASYN) or undersampling was applied, as the dataset is small enough that synthetic oversampling would risk introducing noisy near-duplicate examples in the minority classes. The imbalance ratio of 8.3:1 is well within the validated range for weighted cross-entropy (up to approximately 200:1 in the literature) and does not require specialised techniques beyond class weighting.

2.8 Ethical Considerations

This study was submitted to the KNUST Committee on Human Research, Publications and Ethics (CHRPE) in March 2026 (reference number pending approval). No student interaction log collection or user study data collection commenced before ethics approval. All student participants will provide written informed consent before any interaction logging. Student identities are de-identified via synthetic hash replacement before analysis. The KNUST Multimedia Learning Unit provided written permission for FAISS indexing of official IDL lecture transcripts. Data processing complies with the Ghana Data Protection Act (2012) and applicable GDPR provisions. The Nyansapo-QIC-WCE classifier training dataset is derived from anonymised system query logs and does not contain personally identifiable information.

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